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Nick Petalas

Senior Software Engineer

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EMPLOYMENT

Senior Frontend Developer | Eeze (Game Innovation) — Sep 2025 – Present

- Lead frontend developer across multiple concurrent iGaming projects, building with React, Zustand, Protobuf, and Docker.
- Led end-to-end frontend development of **Operation Tools**, coordinating closely with backend engineers.
- Sole developer and project lead on the **Backoffice Template Service** — designed the architecture, implemented localization, test coverage, and a multi-stage Docker deployment pipeline. The service became a reusable platform adopted across multiple products.
- Co-delivered the **MMD** and **M12D** iGaming projects, integrating template service support and contributing through to launch.

Founder & Lead Engineer | DailyGoal.fit — Oct 2024 – Present

- Founded and shipped a **health and fitness tracking platform** across web and iOS (live on the App Store).
- Designed and implemented full-stack architecture using Next.js, React Native (Expo), Convex, and Vercel.
- Shipped an iOS app featuring nutrition tracking, barcode scanning (USDA FoodData Central & OpenFoodFacts), workout logging, custom goals, Apple Health integration, streak systems, and in-app subscriptions.
- Designed real-time database schema to support efficient querying, user-specific goal logic, streak tracking, and historical analytics.
- Developed CI/CD workflows and infrastructure automation including automated TestFlight builds and deployments.

Senior Software Engineer | Genesis Global Technology LTD — Dec 2019 – Oct 2024

- **Technical Lead** for the "Learn" team and core member of the web platform team over a 5-year tenure.
- Led the modernization of the web platform across three generations — migrating from legacy Angular to StencilJS, then to Microsoft FAST with custom web components and a bespoke design system.
- Collaborated with client teams on React integrations, bridging platform and product engineering.
- Mentored developers, onboarded new starters, and resolved tier 3 escalation issues across the platform.
- Built developer tooling including **Genx CLI**, low-code scaffolding, app seeds, and custom Docusaurus plugins.
- Owned DevOps responsibilities including AWS Amplify, GitHub Actions pipelines, and Infrastructure as Code.

Software Engineer | Elemental Concept LTD — Sep 2018 – Dec 2019

- Full-stack engineer delivering across mobile, web, and backend on four concurrent client projects.
- Led major refactoring and microservice extraction on **Gift & Go**, a B2B rewards platform (Java 11, Spring Boot, Angular, AWS, Docker) — built APIs for enterprise clients including Scientific Games.
- Shipped **CorResilience** (health coaching app with lab and Fitbit integration) and **WellteQ** (wellness platform — Ionic, Cordova, Node.js, Java, MySQL, MongoDB).
- Built **Go Fast**, an internal rapid prototyping toolset (React, Angular, NestJS).

Software Engineer | Enigma Software Solutions LTD — Feb 2017 – Sep 2018

- Built and maintained iGaming platform tools for online betting clients: client portals, payment systems, trader and bet management, reporting, and retail cashier systems.
- Developed and maintained Java/SQL back-end services across legacy (PrimeFaces, Oracle WebLogic) and modern stacks.

Software Engineer | Safemarket LTD Dec 2014 – Oct 2016

- Designed and developed internal tools and knowledge extraction systems for the **Erybo** project (Java, SQL, RDF, Jsoup, Selenium, Apache Jena).
- Built automated media processing tools (Java, ffmpeg), trivia content generators, and entity parsers.
- Developed backend services and features for the **Quizedia** Android app.
- Trained and supervised interns in scraping and quiz generation.

EDUCATION

University of Macedonia | Thessaloniki, Greece Oct 2012 – Oct 2015

BSc In Applied Informatics. Final Grade: 7.7 / 10
Thesis: A study on Optimization of Object-Oriented Design using Genetic Algorithms.
Grade: 10 / 10; Supervisor: Alexander Chatzigeorgiou.

OTHER TECHNICAL EXPERIENCE

- **Homelab:** Built and maintained a Proxmox-based homelab running multiple VMs, including:
  - TrueNAS VM with HBA passthrough for an 8-drive RAIDZ2 NAS.
  - Debian VM running dockerized services: Traefik, Plex, Radarr, Sonarr, Lidarr, SABnzbd, qBittorrent, Filebrowser, Glances, Netdata, Home Assistant, and more.
  - Managed network stack: OPNsense router, Pi-hole for DNS-level adblocking, Unbound as recursive resolver, and Omada SDN Controller for access point management.
- **Heuristic Vector Image Generator:** Genetic algorithm that evolves layered semi-transparent polygons to approximate a target image. Built standalone versions in Java (multithreaded) and Rust (GPU-accelerated evolution via wgpu compute shaders, portable SIMD, Leptos/WASM real-time viewer). Also shipped a web version at artgen.xyz using StencilJS, Rust/WASM, and Firebase.
- **Custom Database Merger:** Built a system to augment a large RDF database (based on Freebase) by scraping websites, performing entity recognition (OpenNLP), and fact extrapolation. (Java, MySQL, RDF)
- **BTC Risk Metric:** Developed a custom Bitcoin risk oscillator, combining metrics like EMA ratios, RSI, Mayer Multiple, Puell Multiple, and others to assess cyclical risk. Published as a TradingView indicator.
- **Auto Conf Scripts:** Maintained a cross-platform setup automation suite for Linux/macOS. Installs hundreds of dependencies, patches fonts, configures zsh, clones repos, and links dotfiles to streamline new machine provisioning.

PROGRAMMING LANGUAGES

- **Primary focus:** TypeScript / JavaScript (React, React Native, Node.js, and experience with multiple other frameworks and native web components), HTML, CSS, Bash (scripting), Dockerfiles.
- **Previously used professionally:** Java, Kotlin, Python, SQL.
- **Limited experience:** C, C++, R, PHP, MATLAB, SPARQL.
- **Currently learning:** Rust and Elixir.